

4.d) $y_{k+3} + y_{k+2} - 5y_{k+1} + 3y_k = 0$, sujeta a $y_0 = 0$, $y_1 = -1$, $y_2 = 2$

APLICAMOS TRANSFORMADA $\mathcal{Z}$

$$\mathcal{Z}\{y_{k+3}\} + \mathcal{Z}\{y_{k+2}\} - 5\mathcal{Z}\{y_{k+1}\} + 3\mathcal{Z}\{y_k\} = 0$$

$$z^3 Y(z) - \cancel{z^3 y_0}^0 - \cancel{z^2 y_1}^0 - \cancel{z y_2}^0 + \cancel{z^2 y_0}^0 - \cancel{z y_1}^0 - 5z Y(z) - \cancel{2 y_0}^0 + 3 Y(z) = 0$$

$+z^2$ $-2z$ $+z$

$$z^3 Y(z) + z^2 - 2z + z^2 Y(z) + z - 5z Y(z) + 3 Y(z) = 0$$

$$(z^3 + z^2 - 5z + 3) Y(z) = -z^2 + z$$

$$Y(z) = \frac{-z^2 + z}{(z^3 + z^2 - 5z + 3)}$$

$$-z^2 + z = -z(z-1)$$

$$Y(z) = \frac{-z^2 + z}{(z-1)^2(z+3)}$$

$$Y(z) = \frac{-z(z-1)}{(z-1)^2(z+3)}$$

$$\frac{Y(z)}{z} = - \frac{1}{(z-1)(z+3)}$$

$$\frac{Y(z)}{z} = -\frac{1}{4} \frac{1}{z-1} + \frac{1}{4} \frac{1}{z+3}$$

$$\mathcal{Z}^{-1} \left[Y(z) = -\frac{1}{4} \frac{z}{z-1} + \frac{1}{4} \frac{z}{z+3} \right]$$

$$y_k = -\frac{1}{4}(1)^k + \frac{1}{4}(-3)^k$$

$$y_k = \frac{(-3)^k - 1}{4} // ; k \geq 0$$

$$\begin{array}{r|rrrr} z^3 + z^2 - 5z + 3 & 1 & 1 & -5 & 3 \\ & & 1 & 2 & -3 \\ \hline & 1 & 2 & -3 & 0 \end{array}$$

$z^2 + 2z - 3 = (z+3)(z-1)$
 $z + 3 = (z-1)^2(z+3)$
 $z - 1 = (z-1)^2(z+3)$

$$= \frac{A}{z-1} + \frac{B}{z+3}$$

$$A = \frac{-1}{z+3} \Big|_{z=-1}$$

$$A = -\frac{1}{4} //$$

$$B = -\frac{1}{z-1} \Big|_{z=-3}$$

$$B = \frac{1}{4} //$$